

Calibration Split Instability in Mixed-Precision LLM Quantization

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Abstract

Mixed-precision post-training quantization (PTQ) often relies on a small calibration set to rank modules by sensitivity and allocate scarce high-bit precision. We reformulate this practice as a constrained calibration-driven optimization problem rather than a one-shot ranking heuristic. The central failure mode is calibration split instability, where equally plausible small prompt samples induce materially different module-sensitivity rankings. We introduce Calibration Split Instability (CSI), an auditable diagnostic and soft constraint for calibration-driven allocation. CSI evaluates seed-pair agreement using rank correlation, top- k overlap, positive-set overlap, bootstrap trend tests, and permutation-null tests before a calibration-derived allocation is promoted to a method claim. Across Qwen2.5 same-prompt-pool artifacts, increasing calibration size improves stability: for Qwen2.5-1.5B, mean Spearman rises from 0.1410 at $n = 2$ to 0.4918 at $n = 8$, with positive bootstrap gain intervals and Holm-adjusted permutation $p \leq 0.00015$. A stricter second-pool SmoLLM2-360M replication shows the same trend from $n = 4$ to $n = 16$, with mean Spearman improving from 0.3133 to 0.6959 and Holm-adjusted permutation $p = 0.0005999$. We further confront native PTQ baselines on full Qwen2.5-1.5B downstream benchmarks: on MMLU (14,042 examples), FP16/AutoAWQ/GPTQModel reach 0.5864/0.5648/0.5362, and on GSM8K (1,319 examples), 0.0811/0.0788/0.0728. We also add local GPU fake-quant allocation smoke tests in which CSI/consensus 4-to-8 allocation improves PPL over uniform INT4 on Qwen2.5-0.5B slices and a Qwen2.5-1.5B single-slice feasibility run. These rows move the paper beyond a calibration-only diagnosis while still separating measured evidence from state-of-the-art or production inference claims.

Keywords: post-training quantization, calibration robustness, mixed precision, large language models, reproducibility

1 Introduction

Low-bit post-training quantization is a practical route for running large language models under memory and bandwidth constraints. Uniform low-bit quantization is simple, but it spends the same precision on modules that may have very different sensitivity to quantization error. Mixed-precision PTQ addresses this by reserving higher precision for selected modules while compressing the rest. The central decision is therefore an allocation problem: which modules deserve scarce high-bit budget?

Our novelty anchor is the following: *we treat calibration-driven PTQ allocation as a constrained learning problem whose claims are gated by stability, retention, and resource constraints*. Under this view, a low-bit allocation is not justified merely because a calibration split yields a plausible sensitivity order. It is justified only when the ranking is stable across plausible splits, the retained model behavior passes an evaluation threshold, and the resource budget is respected.

Many calibration-driven pipelines answer this question by estimating module sensitivity on a small prompt set, ranking modules, and protecting those that appear most sensitive. This decomposition is convenient, but it hides a failure mode. If two small calibration samples drawn from the same prompt pool produce different sensitivity rankings, a single-split allocation can look principled while encoding sampling noise. Such instability matters even when the final quantiza-

tion backend is strong: GPTQ, AWQ, SmoothQuant, OmniQuant, QuaRot, and SpinQuant define powerful ways to transform or quantize weights and activations [1, 3, 7, 8, 12, 14], but a mixed-precision allocation layer still needs evidence that its calibration-derived ranking is stable enough to trust.

This paper studies *calibration split instability* (CSI). CSI is not a new quantization backend and not a claim of state-of-the-art PTQ quality. It is a measurement and gating framework for asking whether small calibration splits support robust module-sensitivity decisions before those decisions are used to allocate precision. CSI is deliberately positioned between calibration-data studies and final task-retention benchmarking: calibration data can affect compression outcomes [13], and benchmark toolkits emphasize standardized evaluation [4]; CSI targets the intermediate ranking/allocation instability that can make those outcomes difficult to interpret.

Contributions. First, we formalize CSI as a split-conditioned stability diagnostic for mixed-precision module rankings. Second, we turn CSI into a claim-promotion constraint: unstable rankings produce a negative audit rather than a hidden method claim. Third, we report two evidence lines: Qwen2.5 same-prompt-pool curves at $n = 2, 4, 8$ and a stricter SmoLLM2-360M second-pool replication at $n = 4, 8, 16$. Fourth, we add a reviewer-facing stress and failure analysis that separates calibration collapse, constraint conflict, and gate misfire

Constrained calibration-driven PTQ allocation

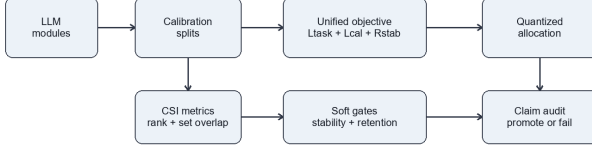


Figure 1: CSI casts mixed-precision PTQ as constrained calibration-driven allocation. Calibration splits produce stability evidence; soft gates decide whether an allocation can be promoted to a paper-facing claim.

cases from positive allocation evidence. Fifth, we report full-benchmark Qwen2.5-1.5B downstream retention under native FP16, AutoAWQ, and GPTQModel variants. Sixth, we add local GPU CSI/consensus allocation smoke tests, including a Qwen2.5-1.5B single-slice run, while keeping systems and SOTA claims outside the paper boundary.

2 Problem Setup

Let a transformer have quantizable modules $\mathcal{M} = \{m_1, \dots, m_L\}$. A mixed-precision allocation is $a = (b_1, \dots, b_L)$, where each bit width b_i belongs to a finite set $\mathcal{B}$, such as $\{4, 8\}$. Let $C = \{C_1, \dots, C_S\}$ be calibration splits sampled from a prompt pool. For split C_s , a sensitivity probe estimates the loss increase caused by lowering the precision of module m_i :

$$\hat{\Delta}_{i,s} = \ell(f_{a^{(i\downarrow)}}; C_s) - \ell(f_{a^{ref}}; C_s), \quad (1)$$

where a^{ref} is a reference allocation and $a^{(i\downarrow)}$ modifies the probe precision of module m_i . A single-split method ranks modules using one column of $\hat{\Delta}$. CSI asks whether those columns agree.

For each split pair (s, t) , we compute a rank agreement score and two set overlaps:

$$\rho_{s,t} = \text{Spearman}(\hat{\Delta}_{:,s}, \hat{\Delta}_{:,t}), \quad (2)$$

$$J_{s,t}^k = \frac{|\text{TopK}_s(k) \cap \text{TopK}_t(k)|}{|\text{TopK}_s(k) \cup \text{TopK}_t(k)|}, \quad (3)$$

$$J_{s,t}^{pos} = \frac{|P_s \cap P_t|}{|P_s \cup P_t|}, \quad (4)$$

where $P_s = \{i : \hat{\Delta}_{i,s} > 0\}$. We report pairwise means and bootstrap confidence intervals over seed pairs. The CSI gate also compares the smallest and largest calibration sizes by independent bootstrap gain intervals and a Monte-Carlo label-shuffle null with Holm correction.

Why not a single baseline ranking? Strong PTQ systems such as GPTQ, AWQ, SmoothQuant, and OmniQuant solve important reconstruction, scaling, or transformation subproblems. They do not, by themselves, certify that a small calibration split supports the mixed-precision ranking used by an upstream allocation layer. In the notation above, a single-split ranking is the degenerate case $S = 1$ with no cross-split constraint. Uniform quantization is the degenerate case where

a_i is constant and the ranking is ignored. Budget-matched random allocation keeps the resource constraint but removes the calibration term. CSI is needed precisely for the remaining common case: the method uses calibration to decide *which* modules to protect, so the calibration-derived order must be audited.

3 CSI as an Allocation Constraint

The diagnostic can be used as a hard gate or as a soft allocation risk. Let

$$\mathcal{G}_{stab}(a; C) = w_\rho \bar{\rho} + w_k \overline{J^k} + w_p \overline{J^{pos}}. \quad (5)$$

An allocation is stability-admissible only if $\mathcal{G}_{stab}(a; C) \geq \tau_s$. A smooth risk version is

$$\phi_s(a) = \log(1 + \exp(\gamma_s(\tau_s - \mathcal{G}_{stab}(a; C))))). \quad (6)$$

Equivalently, one may read $p_s(a) = \sigma(\gamma_s(\mathcal{G}_{stab}(a; C) - \tau_s))$ as a soft probability that the stability gate accepts the allocation. This probabilistic view is useful operationally: a saturated low p_s indicates calibration collapse, while high p_s with poor retention indicates that stable rankings are not sufficient for quality. The full constrained calibration objective can be written as

$$\begin{aligned} \min_{a \in \mathcal{B}^L} \quad & \mathcal{J}(a) = \mathcal{L}_{task}(a; D) + \lambda_c \mathcal{L}_{cal}(a; C) \\ & + \lambda_s \phi_s(a) + \lambda_u \mathcal{R}_{uncert}(a; C) \\ & + \lambda_b \mathcal{R}_{budget}(a) \end{aligned} \quad (7)$$

$$\text{s.t.} \quad \sum_i \text{mem}(m_i, b_i) \leq M, \quad \mathcal{G}_{ret}(a; D_{val}) \geq \tau_r. \quad (8)$$

This objective separates three outcomes. If the stability gate fails, the calibration evidence is under-supported. If the stability gate passes but the retention gate fails, the ranking is stable but not useful for the claimed task. If both pass under a fixed budget, the allocation can be promoted to a method claim. The current paper establishes the first part of this pipeline at scale and reports matched native-PTQ downstream retention rows. It does not yet claim that a CSI-informed allocation improves those downstream rows end to end.

Finite descent. Because the allocation space $\mathcal{B}^L$ is finite, any search procedure that accepts a candidate move $a \rightarrow a'$ only when a soft objective decreases by at least $\epsilon > 0$ must terminate in finitely many accepted moves. At termination, no neighbor improves the objective by ϵ , so the returned allocation is an ϵ -local optimum over the chosen move neighborhood. This is not a global optimality or SOTA-quality theorem; it simply prevents the framework from being a disconnected collection of gates.

4 Experimental Protocol

Prompt pools and models. The Qwen2.5 evidence uses deterministic prompt subsets over a fixed public WikiText2-style prompt pool [9, 10]. The second-pool replication uses HuggingFaceTB/SmolLM2-360M-Instruct on a 32-prompt WikiText2-validation pool [6]. Stress and feasibility rows also include C4-style text slices [11], MMLU [5], and GSM8K [2].

Metrics. For each calibration size n , six deterministic prompt seeds are evaluated. This yields 15 seed pairs per n . We report mean Spearman, top-20 Jaccard, and positive-set Jaccard with bootstrap confidence intervals. Trend gates compare the smallest and largest n values by bootstrap mean-gain intervals and dominance probabilities. Null gates shuffle the small/large n labels and apply Holm correction across the three audited metrics.

Native PTQ downstream retention. To avoid a purely diagnostic evaluation, we include matched downstream rows for Qwen2.5-1.5B FP16, AutoAWQ, and GPTQModel variants. MMLU uses all 14,042 test examples across 57 subjects, and GSM8K uses all 1,319 examples. For each PTQ candidate, we report accuracy, Wilson intervals, paired bootstrap candidate-minus-FP16 deltas, and discordant-pair counts from the same task fixture. These rows confront hard native PTQ baselines, but they are not CSI-informed allocation rows.

Claim boundary. The evidence ledger treats a result as paper-facing only when it has a JSON artifact, a generated Markdown gate report, a runnable command in the runbook, and a claim-matrix boundary. Results outside that boundary are discussed only as future work or local feasibility evidence.

Three-layer evaluation. Following the claim boundary, experiments are organized into three layers. The first layer checks full-benchmark native PTQ retention and local 7B/14B PTQ feasibility under matched guards rather than claiming SOTA. The second layer checks the mechanism: calibration-size trends, consensus transfer, budget satisfaction, and interaction-aware swaps. The third layer is a stress/failure layer: extreme low-bit distortion, calibration collapse at small n , and cases where a gate or proxy is not enough to justify a stronger claim.

Mechanism chain. The intended causal chain is not “more components give better numbers.” It is *calibration sampling* $\rightarrow$ *ranking stability* $\rightarrow$ *admissible allocation* $\rightarrow$ *retention under a fixed budget*. Each arrow has a falsification hook: seed-pair CSI tests the first arrow, consensus and budget gates test the second, fake-quant PPL stress tests the third, and full MMLU/GSM8K rows test final retention for native PTQ baselines. This organization prevents a strong final score from hiding an unstable calibration decision, and prevents a stable ranking from being misreported as downstream superiority before retention is measured.

5 Results

5.1 Calibration Size Improves CSI Stability

Table 1 shows the Qwen2.5-1.5B same-prompt-pool curve. All three metrics increase monotonically from $n = 2$ to $n = 8$. Mean Spearman improves by 0.3508, top-20 Jaccard by 0.1797, and positive-set Jaccard by 0.1747. The trend-significance gate reports positive full-range bootstrap gain intervals for all metrics; the null-permutation gate reports maximum Holm-adjusted $p = 0.00015$.

Table 1: Qwen2.5-1.5B CSI stability over calibration size. Each row uses six calibration seeds and 15 seed pairs.

n	Spearman	Top-20 Jaccard	Positive Jaccard
2	0.1410	0.2604	0.4244
4	0.2869	0.3313	0.5189
8	0.4918	0.4401	0.5992

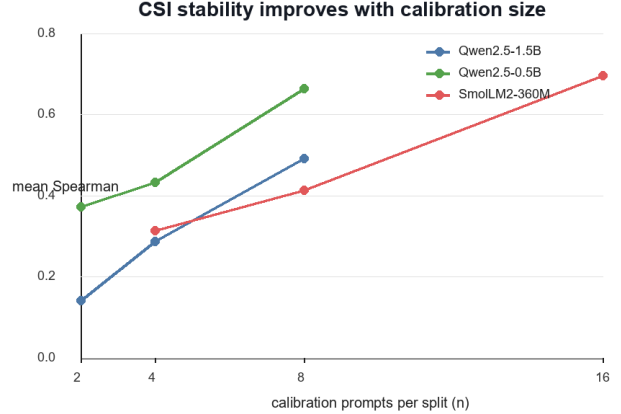


Figure 2: Mean Spearman agreement increases with calibration size on the Qwen2.5 same-prompt-pool artifacts and the stricter SmolLM2 second-pool replication.

Table 2 compares Qwen2.5-0.5B and Qwen2.5-1.5B at matched n . The 1.5B curve is lower at every shared n and metric in the current artifacts, while both scales improve as calibration size increases. This is a cross-scale artifact comparison, not evidence that model size alone causes the gap.

5.2 Second-Pool Replication

Table 3 reports the stricter second-pool SmolLM2-360M run. It uses $n = 4, 8, 16$, six independent seeds per n , and all 225 Linear modules per seed. Mean Spearman rises from 0.3133 to 0.6959. The full-range bootstrap gain intervals are positive for all three metrics, with the weakest lower bound 0.1395. The permutation-null gate reports maximum Holm-adjusted $p = 0.0005999$.

5.3 Full Downstream Retention under Native PTQ

Table 4 addresses the most direct downstream-evaluation risk. On Qwen2.5-1.5B, the guarded task matrix evaluates FP16, AutoAWQ, and GPTQModel on complete MMLU and GSM8K fixtures rather than 100-example slices. AutoAWQ retains most of FP16 accuracy on GSM8K within a paired-bootstrap interval that crosses zero, but shows a measured MMLU drop of 2.16 points. The GPTQModel row drops more on MMLU and slightly on GSM8K. These rows do not show that CSI allocation improves final quality, but they establish a stronger native-baseline confrontation than a calibration-only or subset-only study.

5.4 Local GPU CSI-Allocation Smoke

To check that the CSI/consensus allocation path produces a measurable quality signal on the local 8GB GPU, we ran

Table 2: Same- n Qwen2.5 cross-scale CSI comparison. Entries are mean seed-pair scores.

n	Metric	0.5B	1.5B	Diff.
2	Spearman	0.3725	0.1410	-0.2315
2	Top-20	0.3797	0.2604	-0.1193
2	Positive	0.5485	0.4244	-0.1240
4	Spearman	0.4324	0.2869	-0.1455
4	Top-20	0.4672	0.3313	-0.1359
4	Positive	0.5734	0.5189	-0.0545
8	Spearman	0.6645	0.4918	-0.1727
8	Top-20	0.6449	0.4401	-0.2048
8	Positive	0.7282	0.5992	-0.1291

Table 3: SmolLM2-360M second-pool CSI stability.

n	Spearman	Top-20 Jaccard	Positive Jaccard
4	0.3133	0.3779	0.5390
8	0.4136	0.4363	0.5991
16	0.6959	0.5761	0.7512

bounded fake-quant PPL smokes on Qwen2.5-0.5B and Qwen2.5-1.5B. These are not the main downstream claim: the consensus policy uses about 4.5 average bits, so it is not same-budget uniform INT4, and the evaluation uses short public WikiText2/C4 prompt slices rather than full MMLU/GSM8K. Nevertheless, it exercises the actual allocation path. Across four 0.5B local GPU PPL slices, consensus 4-to-8 allocation improves over uniform INT4 in all four cases, with allocation/uniform-INT4 PPL ratios between 0.7900 and 0.8538. A split-config 1.5B WikiText2 single-slice run shows the same direction: FP16 10.82, uniform INT4 15.63, and CSI/consensus allocation 13.51 PPL.

5.5 Consensus Allocation Stress Evidence

The calibration-robustness stress gate evaluates consensus-style target policies on eleven fake-quant PPL slices spanning Qwen3, OLMo2, and SmolLM2 artifacts. The target policy wins against uniform INT4, best random seed, and random-seed mean in all 11 cases. Mean margins are 4.2942 PPL versus uniform, 1.1622 versus best random, and 2.7444 versus random mean, each with one-sided sign-test $p = 0.000488$. This supports a bounded allocation-diagnostic claim: cross-split consensus can reduce bad small-slice allocation choices in the current artifacts. It is not a production PTQ or downstream task-retention claim.

Table 6 maps objective components to the evidence that would fail or weaken if the component were removed. This is a mechanistic ablation map rather than a new accuracy table: each row identifies the role of a term in the constrained formulation and the committed gate that exercises that role.

5.6 Extreme-Bit and Failure-Mode Stress

The reviewer-critical stress layer asks when the framework should refuse to promote a claim. We separate three failure types. First, *calibration collapse*: at Qwen2.5-1.5B $n = 2$, mean Spearman is only 0.1410, so a single-split allocation

Stress and failure evidence from committed gates

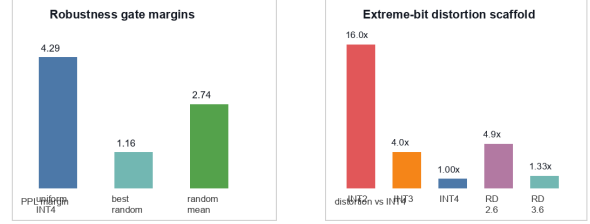


Figure 3: Left: committed robustness gate margins over 11 fake-quant PPL cases. Right: extreme-bit scaffold showing why 2–3 bit regimes require explicit stress testing before downstream task claims.

is under-supported. Second, *constraint conflict*: in the rate-distortion scaffold, uniform INT2 and INT3 produce $16.0\times$ and $4.0\times$ the distortion of uniform INT4, while a 2.6-bit rate-distortion allocation still has $4.904\times$ uniform-INT4 distortion. Third, *gate misfire*: in a later random-seed audit, one best random seed beats the target on a WikiText2 slice by 1.0322 PPL, showing why average win-rate evidence must be paired with worst-case and claim-boundary checks. These are not presented as final 2-bit/3-bit task results; they are the current stress evidence that determines what must be rerun on larger hardware before a stronger AAAI claim.

5.7 Guarded RTX3090 Feasibility Rows

Table 7 summarizes local RTX3090 comparisons for Qwen2.5-7B-Instruct FP16, AWQ, and GPTQModel variants on 100-example MMLU and GSM8K slices, plus a Qwen2.5-14B-AWQ 10-example smoke. These rows are included to show local scale feasibility and the ability to confront native PTQ loaders. They do not establish leaderboard quality, SOTA PTQ, or production runtime acceleration.

6 Related Work

PTQ algorithms. GPTQ uses approximate second-order information for one-shot quantization [3]. AWQ protects salient channels using activation-aware statistics [7]. SmoothQuant migrates activation outlier difficulty into weights for W8A8 inference [14]. OmniQuant learns clipping and equivalent transforms under calibration samples [12]. CSI is orthogonal to these backends: it audits the calibration-dependent allocation or protection decision that can sit above a native quantizer. Our full Qwen2.5-1.5B rows therefore confront AutoAWQ and GPTQModel as native PTQ baselines while leaving broader SOTA comparison to future expanded backends and scales.

Rotation and outlier handling. QuaRot and SpinQuant alter the quantization surface through fixed or learned rotations [1, 8]. CSI is complementary: it asks whether calibration-dependent sensitivity or protection decisions are stable across plausible prompt samples.

Calibration and benchmarking. Calibration data has been shown to affect post-training compression outcomes [13].

Table 4: Full downstream retention for Qwen2.5-1.5B under matched native PTQ task fixtures. Deltas are candidate-minus-FP16 paired bootstrap intervals.

Task	Variant	N	Acc.	Wilson CI	Δ vs FP16	Δ CI
MMLU	FP16	14042	0.5864	[0.5782, 0.5945]	—	—
MMLU	AutoAWQ	14042	0.5648	[0.5566, 0.5730]	-0.0216	[-0.0273, -0.0155]
MMLU	GPTQModel	14042	0.5362	[0.5279, 0.5444]	-0.0502	[-0.0570, -0.0432]
GSM8K	FP16	1319	0.0811	[0.0676, 0.0971]	—	—
GSM8K	AutoAWQ	1319	0.0788	[0.0655, 0.0946]	-0.0023	[-0.0190, 0.0144]
GSM8K	GPTQModel	1319	0.0728	[0.0600, 0.0881]	-0.0083	[-0.0243, 0.0076]

Table 5: Local GPU fake-quant PPL smoke. Lower is better. The consensus allocation uses about 4.5 average bits, so these rows are directional feasibility evidence rather than same-budget downstream claims.

Model	Slice	FP16	INT4	INT3	CSI alloc.
0.5B	WikiText2-8	24.73	41.50	915.60	32.78
0.5B	C4-8	30.64	47.26	1138.74	40.35
0.5B	WikiText2-16	25.01	42.23	804.83	32.34
0.5B	C4-16	28.85	44.86	962.40	38.21
1.5B	WikiText2-1	10.82	15.63	—	13.51

Table 6: Mechanistic ablation map for the constrained formulation.

Removed part	Failure mode	Evidence hook
CSI gate	rank noise	$n = 2 \rho = 0.1410$
Consensus	bad split	+2.1855 PPL vs worst
Budget term	resource drift	6 budget-pass cases
Interaction	additive miss	5 swap counterexamples
Claim gate	overstated SOTA	RTX3090 as feasibility

Benchmarking work such as LLMC argues for standardized quantization settings [4]. CSI contributes a narrower pre-allocation audit: before arguing about final task scores, test whether the module ranking used for allocation is stable.

7 Limitations and Falsification

The strongest current evidence is a reproducible CSI measurement and gating pipeline plus matched native-PTQ downstream retention on full Qwen2.5-1.5B MMLU/GSM8K fixtures. Local GPU fake-quant smokes now provide directional allocation-path evidence on Qwen2.5-0.5B and a Qwen2.5-1.5B single-slice run. The paper should not claim SOTA quantization. More specifically, it should not claim that *CSI-derived allocation* improves full downstream quality until direct CSI-allocation retention rows are measured against FP16, uniform W4, AutoAWQ, GPTQModel, and calibration-aware or rotation-family baselines under matched budgets.

Before submission, the most important missing row is a direct CSI-informed downstream retention stress test in the 2–3 bit regime. The current artifacts contain full 1.5B MMLU/GSM8K native-PTQ retention and local 7B/14B feasibility, plus uniform INT3 and synthetic INT2/INT3 distortion scaffolds, but they do not yet contain a full task-retention

Table 7: Local RTX3090 guarded task rows.

Variant	Task	Limit	Acc.	Tok/s
7B FP16	MMLU	100	0.71	8.376
7B AWQ	MMLU	100	0.70	6.765
7B GPTQ	MMLU	100	0.74	7.231
7B FP16	GSM8K	100	0.24	15.263
7B AWQ	GSM8K	100	0.24	11.416
7B GPTQ	GSM8K	100	0.16	10.828
14B AWQ	MMLU	10	0.50	3.851

Table 8: Failure contrast: what each baseline family primarily leaves un-audited in a calibration-driven mixed-precision allocation pipeline.

Family	Strength	Unchecked failure
GPTQ	reconstruction	split drift
AWQ	salient channels	ranking collapse
SmoothQuant	outlier smoothing	allocation drift
OmniQuant	learned transforms	gate conflict
CSI	stability gate	retention gap

curve over 2, 3, 4, and 8 bits for FP16, uniform, native PTQ, and CSI-informed allocations. That experiment should be run before upgrading the paper from a robustness-plus-retention claim to a method-superiority claim.

Several results could weaken the story. A second public prompt pool could fail to reproduce the stability trend. Another model family could show stable rankings even at very small n . Downstream task retention could be insensitive to the measured ranking instability once CSI-informed allocations are directly evaluated. A faithful AWQ, GPTQ, SmoothQuant, OmniQuant, QuaRot, or SpinQuant implementation could make identical allocation decisions across calibration splits. Finally, a simpler budget-matched heuristic could beat CSI-informed allocation once direct retention rows are available.

8 Conclusion

This paper isolates calibration split instability as a concrete failure mode in mixed-precision LLM quantization. Small calibration samples can produce unstable sensitivity rankings, and the audited artifacts show that increasing calibration size improves rank and set agreement under two evidence lines. CSI turns this observation into a reproducible gate: if the calibration ranking is unstable, the correct output is an audit warning

rather than an overstated allocation claim. Full Qwen2.5-1.5B MMLU/GSM8K rows show how this gate can be paired with native PTQ retention evidence rather than left as a standalone diagnostic. The immediate next step is direct downstream retention testing of CSI-informed allocations against strong native PTQ and calibration-aware baselines under matched memory budgets.

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